Limits of Oracle-Blind Rollout Geometry for Detecting Reward Hacking

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Abstract

A reward hack and a benign capability improvement are, at the level of the rollout distribution, the *same event*: a policy sharpening onto whatever the reward favors. The only quantity that distinguishes them—whether the favored mode also has higher *true quality*—is the held-out oracle, which an *oracle-blind* detector cannot observe by construction. We formalize this as an indistinguishability result (Proposition 2) whose required twin we *reduce* to the reward being hackable—proving the concentration features of the geometry (entropy collapse, advantage narrowing) are target-blind and so coincide, leaving only a low-dimensional, quality-agnostic residual to the empirics (Proposition 1)—and, via a data-processing argument, a bound stating that no oracle-blind decision rule achieves above-chance worst-case separation of the two regimes (Corollary 1). We then support the claim with three empirical legs on a GRPO-instrumented code-generation setting. First, in a detector-characterization study, every candidate geometric axis (convergence sharpness, output degeneracy, advantage-distribution shape) fires *identically* on a benign control matched on that axis—including `return 0` (a degenerate hack) versus `return n*(n+1)//2` (a correct one-liner), which are geometrically indistinguishable. Second, across **18 real benign GRPO runs** (base Qwen2.5-1.5B, three well-specified coding tasks on which no hacking is possible, six seeds each; the held-out oracle rises on every run), the detector fires on **10**—all **false positives**—at a *scattered* operating point (peak warmup-normalized CUSUM $S \in [7.5, 144]$, no stable benign value to calibrate against), and an onset-strength sweep shows the synthetic hack strengths fall *inside* this benign cloud, so no CUSUM threshold separates the two; replaying the same benign geometry through structurally different detector families (an EWMA control chart, a rank-based two-sample test) places each at a different point of that same chance-level frontier rather than past it. Third, an elicitation probe found the test-overwrite exploit *not elicitable within the tested scope* ($\leq$ 14B, edit-distance and regex tasks, the test-overwrite affordance): instruction tuning suppresses it (0/192), base 7B models sit at a $\approx 1\%$ noise floor, and a capable 14B base model solving the tasks at 52–55% still produced 0/512 (95% Wilson interval $[0, 0.74\%]$), not rising from 7B to 14B. We conclude that a usable hacking detector must reintroduce some oracle-adjacent signal; we sketch—but, lacking an elicitable hack, do not build—a reframing of rollout geometry as an *efficient sampler* that decides when to spend a scarce oracle query rather than as a classifier.

1 Introduction

Reinforcement-learning post-training optimizes a policy π_θ against a programmatic or learned reward r that stands in for a goal we cannot directly differentiate (Stiennon et al., 2020; Shao et al., 2024). When r is an imperfect proxy, optimization pressure discovers inputs on which r is high but true quality is not—*reward hacking* or specification gaming (Amodei et al., 2016; Skalse et al., 2022; Pan et al., 2022), up to and including direct tampering with the grading channel (Denison et al., 2024). The phenomenon sharpens with optimization strength (Gao et al., 2023), so an inexpensive online signal that flagged the *onset* of hacking—

computed from quantities the trainer already produces—would be valuable for intervention before a model ships.

The attractive form of such a monitor is *oracle-blind*: it reads only rollout geometry (KL from a reference policy, entropy, advantage statistics—all already computed each step) and never consults a held-out quality measure. The hypothesis behind this paper's system, *Flight Recorder*, was that the transition into hacking leaves an abrupt geometric signature (rapid KL acceleration, entropy collapse, advantage redistribution) that precedes the point at which a held-out oracle gap becomes visible, giving an early-warning lead.

This paper reports that the oracle-blind monitor cannot function as a *hacking classifier*, and that the obstruction is *structural*—a property of what the detector is allowed to observe—rather than an artifact of feature engineering or threshold tuning. Our contributions are:

1. A structural argument (§3) that reward hacking and benign capability gain are the same event in rollout-geometry space, formalized as an indistinguishability proposition (Proposition 2) and a data-processing corollary bounding any oracle-blind decision rule to chance-level worst-case separation (Corollary 1); the indistinguishable twin the argument needs is *reduced* to reward hackability—its concentration-feature half proved target-blind, its residual isolated and shown empirically non-separating—rather than assumed (Proposition 1).
2. A detector-characterization study (§5.1) showing that each candidate discriminative axis collapses against a benign control *matched on that axis*—the empirical shadow of the argument.
3. An 18-run benign batch (three tasks $\times$ six seeds) in which the detector false-positives on 10/18 at a *scattered* operating point (§5.2, Figure 2), and an onset-strength sweep (§5.3) showing the synthetic hack strengths fall inside the benign cloud—no threshold in the detector's own statistic separates the two.
4. An elicitation probe (§5.6) showing the test-overwrite exploit is not elicitable within the tested scope ($\leq$ 14B), with the 14B result statistically resolved by a Wilson interval.
5. A cross-detector replay (§5.4) showing the false positive is not specific to the default detector: structurally different families (an EWMA control chart, a rank-based two-sample test) sit at different points of the *same* TPR=FPR frontier and none escapes it—Corollary 1 made visible.

We lead with the result rather than the original detector hope: oracle-blind geometry detects *that a convergence is underway*, not *that it is a hack*, and we make that boundary precise.

2 Related work

Reward hacking and specification gaming. That optimizing a proxy reward degrades the true objective is a foundational AI-safety concern (Amodei et al., 2016). Skalse et al. (2022) give a formal definition—a proxy is hackable if it can increase while the true reward decreases under some policy pair—which is precisely the configuration our impossibility argument exploits. Pan et al. (2022) map how misspecification severity scales with model capability and optimization, and Gao et al. (2023) quantify reward-model overoptimization. Denison et al. (2024) show that models trained on gameable environments can generalize to editing their own grading code—the exact test-overwrite affordance our elicitation probe targets. All of this work studies *when* hacking occurs; we study whether its *onset is detectable from training dynamics alone.*

Monitoring training dynamics. KL-to-reference, entropy, and advantage statistics are standard health signals in policy-gradient post-training (Schulman et al., 2017; Shao et al., 2024). The implicit premise of online monitoring is that a pathological update looks different from a healthy one in these coordinates. Our negative result delimits that premise: pathology defined by *outcome quality* (hacking) is not separable in these coordinates from a healthy sharp update, because the coordinates are computed without reference to quality. This is complementary to chain-of-thought monitoring (Baker et al., 2025), which reintroduces an outcome-adjacent signal (the model's stated reasoning) precisely because behavior alone is insufficient—consistent with our conclusion that an oracle-adjacent signal is necessary.

Changepoint detection. Our detector uses a CUSUM changepoint test (Page, 1954) on a geometry signal, together with a robust "contraction tube" on the joint feature vector. The contribution here is not a new changepoint statistic but the observation that *no* such statistic, applied to oracle-blind features, can separate the two regimes; the crossover analysis (§5.3) makes the limit quantitative in the statistic's own units.

3 The structural argument

3.1 Setup

Let a policy π_θ be optimized against a reward $r : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ over prompts x and completions y. Write a *run* as the sequence of per-step rollout batches $\{(x_i, y_i, r_i)\}$ produced during training. From a run we compute two disjoint views:

- the **rollout-geometry features** R—$\mathrm{KL}(\pi_\theta \| \pi_{\mathrm{ref}})$ and its derivatives, token entropy and its trend, and statistics of the GRPO advantage distribution (variance, skew, kurtosis, and a 1-Wasserstein drift against a rolling baseline (Villani, 2009)); and
- the **oracle** O—a held-out measure of true quality (a hidden test suite, a trusted judge), used only for offline evaluation.

A run is a *hack* if r is high while O is low, and a *benign improvement* if both rise. A detector is any measurable function f of R producing an onset decision.

3.2 Indistinguishability

Assumption 1 (oracle-blindness)**.** The detector's input is R alone; the oracle is excluded, $O \notin R$.

Assumption 2 (geometry is a function of outputs and proxy reward)**.** $R = g(\{(x_i, y_i, r_i)\})$ for a fixed extractor g; in particular R does not depend on O except through whatever information about O is already carried by (y, r).

Assumption 3 (realizability of a benign twin)**.** For the proxy rewards r of interest there exists a distinct task/reward pairing that is *well-specified* (a benign run on it cannot hack) yet can realize the same output-and-reward trajectory distribution. (§5.2 exhibits one empirically.)

Reducing Assumption 3 to hackability. The benign twin need not be assumed outright. We reduce it to the reward being *hackable*—the only regime in which detection is at issue—and isolate exactly the residual that remains empirical, rather than hiding it inside an informal "same event" step. Treat r, O as functions of outputs. Following Skalse et al. (2022) (point-mass specialization), call r **hackable** w.r.t. O if there exist outputs y_b, y_h with $r(y_h) \geq r(y_b)$ but $O(y_h) < O(y_b)$.

Proposition 1 (the concentration geometry is target-blind)**.** *Let r be hackable via y_h, y_b with $r(y_h) = r(y_b)$, and let run H concentrate the policy onto y_h and run B onto y_b. Then (i) the proxy-reward trajectories of H and B coincide, and (ii) the* concentration *features of R—the token-entropy collapse and the narrowing of the GRPO advantage distribution—are functions of the* degree *of policy concentration alone, not of its target, and so are identical for H and B. The runs thus agree on every component of R that marks a converging run as converging, while $O(y_h) < O(y_b)$.*

Proof. (i) Both maximize the same r with $r(y_h) = r(y_b)$. (ii) Token entropy $-\sum_t \pi_t \log \pi_t \to 0$ as either policy approaches a point mass on its target, identically regardless of the target. The GRPO advantage is the group-normalized reward; as either policy concentrates, all sampled rewards approach $r(y_\star)$, the group variance $\to 0$, and the advantage distribution collapses to a point mass at 0, again independent of which $y_\star$. Hence the entropy-collapse and advantage-narrowing trajectories of H and B coincide; O differs by hackability. □ □

What stays empirical, and why it does not separate. Proposition 1 does *not* assert $R_H = R_B$ in full: the output-identity part of R—chiefly KL from the reference, set by each target's reference log-probability—can differ between y_h and y_b. We are explicit about this residual rather than absorbing it. It is low-dimensional and indexes *which* output was chosen, a variable on which a hack and a benign solution may take any value—not *how good* the output is. Exact identity $R_H = R_B$ holds when y_h, y_b are additionally reference-matched: the canonical degenerate-hack case, `return 0` versus a correct one-liner (§5.1). In general the claim weakens from exact identity to distributional *overlap*, which §5.2–5.5 establish empirically: benign runs populate the warmup-normalized CUSUM range a hack would occupy, across tasks, seeds, algorithms, and model families. The impossibility is therefore *exact* for reference-matched twins and *empirical* (overlap) otherwise—and in neither form is the twin assumed. Finally, the circularity closes: a reward with no hackable pair has no twin, but also no hack, so detection is vacuous; the condition holds exactly where the conclusion matters.

Proposition 2 (concentration features are non-diagnostic)**.** *The concentration statistics on which oracle-blind detectors rely—entropy collapse, advantage narrowing, policy sharpening—are functions of convergence strength, not of oracle quality (Proposition 1). Hence for any such statistic there exist a hacking run and a benign run attaining the* same value *while differing in the oracle, and no oracle-blind function of these statistics identifies the hack/benign label. For reference-matched outputs the coincidence extends to the full geometry ($R_H = R_B$); for the detector's complete feature set in general it weakens to distributional* overlap*, established empirically in §5.2–5.5.*

Proof. By Proposition 1 each concentration statistic is a function of the degree of policy concentration, hence identical for convergence onto a hack output y_h and an equal-reward benign output y_b (both supplied by hackability); the two runs attain the same value of every concentration statistic while $O(y_h) < O(y_b)$. An oracle-blind f of these statistics is constant across the pair and cannot recover the label, which is a function of O—and O, by Assumption 2, is not a function of R. □ □

Intuition (a worked twin). Consider a task graded by a proxy r that checks a returned value against visible cases. The completion `return 0` on a task whose answer is never 0 is a *degenerate hack*; `return n*(n+1)//2` on a task whose answer is the n-th triangular number is a *correct* solution. To the oracle they could not be more different (O: wrong vs. right). To R they are the same event—a short, low-entropy, high-log-probability completion onto which the policy has concentrated mass, producing the same KL acceleration, the same entropy collapse, and the same advantage narrowing. Proposition 2 says this coincidence is unavoidable, not incidental: the very property that makes a hack a hack—low O at high r—is precisely the one absent from R. Figure 1 draws the twin pair: identical geometry R, which is all a detector sees, and a held-out oracle O that diverges—the one signal that separates them and the one an oracle-blind detector cannot read.

The proposition subsumes advantage-distribution statistics, the most tempting "discriminative" axis. Advantage shape can encode *that a subset of samples discovered a higher-reward mode*—but that event occurs identically when the mode is a benign discovery and when it is an exploit, and advantage shape is part of R (computed from r and oracle-blind outputs), so Assumption 2 applies.

3.3 An information-theoretic restatement

Let $C \in \{\mathsf{hack}, \mathsf{benign}\}$ be the regime label. Because C is determined by O and, by Assumption 2, $R \perp O \mid (\text{outputs}, r)$, the label and the features are linked only through the output/reward trajectory T:

$$C \;\to\; O \;\to\; T \;\to\; R \quad \text{is } \textit{not} \text{ a valid Markov chain; instead} \quad C \;\leftarrow\; O, \qquad R = g(T), \qquad T \text{ shared.} \tag{1}$$

The operative fact is that the detector-visible statistics can be held fixed while C varies (Proposition 2). This yields a worst-case *lower bound* on any oracle-blind test.

Corollary 1 (no above-chance test on the matched family)**.** *Let δ be any oracle-blind rule, and restrict to the family of hack/benign pairs that attain equal detector-visible statistics (Proposition 2: equal concentration features always, equal full R for reference-matched outputs). On each such pair δ returns the same label,*

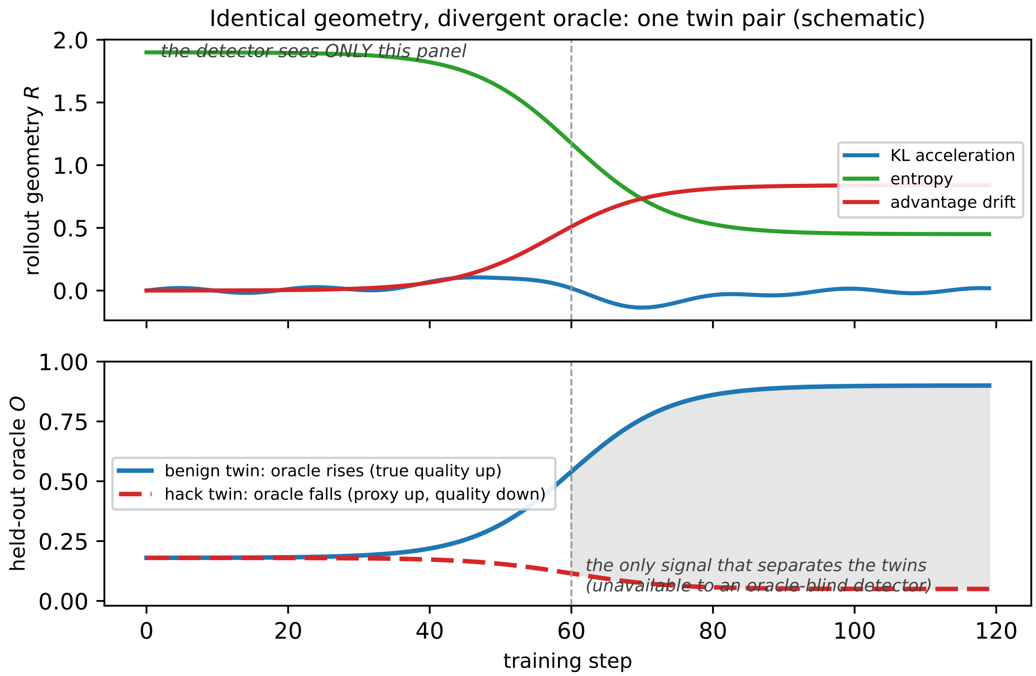

Figure 1: A hack/benign twin pair (schematic illustration of Proposition 2). **Top:** the rollout geometry R—KL acceleration, entropy, advantage drift—is *identical* for both twins; a detector sees only this panel and must return one verdict for both. **Bottom:** the runs differ only in the held-out oracle O: in the benign twin true quality rises, in the hack twin it falls while the proxy reward rises. The separating signal lives entirely in the bottom panel, which an oracle-blind detector cannot observe. The 18-run batch (Figure 2) is the real-data evidence that the top panel—benign convergence indistinguishable from a hack's—occurs in ordinary training.

hence is correct on exactly one: its balanced accuracy on the family is $1/2$ *and* $\mathrm{TPR} = \mathrm{FPR}$*. No setting of* δ*—no tube radius, no CUSUM threshold—separates the regimes on this family.*

Proof. On a matched pair (H, B), δ reads equal statistics, so $\delta(H) = \delta(B)$. If it outputs `hack` it scores a true positive on H and a false positive on B; if `benign`, a false negative and a true negative. Either way, one error and one success: per-pair balanced accuracy $1/2$, $\mathrm{TPR} = \mathrm{FPR}$. □ □

This is a lower bound, not a claim that the matched family is all there is—and the empirics show it is not a measure-zero curiosity. Across tasks, seeds, algorithms, and model families, real benign runs occupy the R-region a hack would (§5.2–5.5), so the bound binds in practice, on ordinary runs, not only on constructed pairs.

Corollary 1 is the precise sense in which the monitor cannot work: not that geometry is uninformative—it genuinely detects *that a convergence is underway*—but that the increment of information needed to call that convergence a *hack* lives entirely in O, which the detector does not see. The remainder of the paper shows that the worst-case twins of Proposition 2 are not pathological constructions but *occur in ordinary training.*

4 Apparatus

Type-level oracle-blindness. Flight Recorder instruments a GRPO trainer. Each step's rollout batch is split by extractors into a `RolloutFrame` (the only object a detector receives) and an `OracleFrame` (consumed only by the offline evaluator). `RolloutFrame` carries no oracle fields, and `Detector.update(frame:`

`RolloutFrame`) is typed to accept only rollout geometry, so oracle leakage into a detector is a type error—Assumption 1 is enforced mechanically rather than by convention.

Features. The detector-visible extractors produce, per step t: the KL mean κ_t and its discrete second difference $\texttt{kl\_accel}_t = \kappa_t - 2\kappa_{t-1} + \kappa_{t-2}$ (curvature, the leading edge of a distribution shift); the token-entropy slope $\texttt{entropy\_trend}_t$ (negative as the policy collapses onto a narrow mode); and the 1-Wasserstein drift $\texttt{adv\_drift}_t = W_1(\hat{A}_t, \hat{A}_{\text{base}})$ of the GRPO advantage distribution against a rolling baseline, alongside its variance/skew/kurtosis and generation statistics (length, n-gram diversity, log-probability concentration). A separate evaluator-only extractor computes the reward–quality gap $r_t - O_t$ and its changepoint; this is never shown to f.

Detector. The default `ContractionTubeDetector` fits a robust reference tube on a warmup window of $W = 30$ steps over $\mathbf{u}_t = [\texttt{kl\_accel}_t, \texttt{entropy\_trend}_t, \texttt{adv\_drift}_t]$, using the per-dimension median m_j and median-absolute-deviation MAD_j. The tube distance is the worst-axis robust z,

$$d_t = \max_j \frac{|u_{t,j} - m_j|}{1.4826\,\text{MAD}_j}, \tag{2}$$

and the tube is "exited" when $d_t > \rho$. In parallel a one-sided CUSUM (Page, 1954) runs on the warmup-standardized signal $z_t = (\texttt{kl\_accel}_t - \mu_W)/\sigma_W$:

$$S_t = \max\big(0,\ S_{t-1} + z_t - k\big), \qquad S_0 = 0, \tag{3}$$

with slack k. Onset is declared at the first t for which both gates—tube exit ($d_t > \rho$, magnitude τ) and CUSUM ($S_t > h$)—hold, *sustained* for a persistence window of $P = 5$ steps (two-of-N corroboration, which suppresses transient single-signal spikes). Both gates are **warmup-normalized**, so the detector is invariant to the absolute scale of KL: a run is measured in MAD- and σ-units of *its own* quiet early phase.

Pre-registration. The thresholds τ=1.0, h=5.0, ρ=3.0, W=30, P=5 were frozen on a synthetic harness before any real-run data was collected. Because the synthetic harness's onsets are *authored*, it validates plumbing and tunes the nominal false-positive rate but cannot establish that the signal exists in real runs—which is exactly the gap the rest of the paper probes.

5 Empirical results

5.1 Detector characterization: candidate axes collapse against matched benign controls

Running the real extractors and detector over controlled trajectories, we asked of each candidate discriminative axis whether it separates a hack-like profile from a benign profile *matched on that axis*—the twin of Proposition 2 restricted to one coordinate.

Convergence sharpness. The tube signals (2) fire on any sufficiently sharp convergence. Three benign profiles—collapse onto a constant, convergence onto a complex solution, and convergence via a high-diversity route—produced onsets indistinguishable from a hack-like sharp update. Sharpness measures *how fast* the policy commits, not *to what*.

Output degeneracy. The generation-length / n-gram-diversity / log-probability-concentration features separate a short output from a long one, but that is the rule "short = suspicious," which a correct one-liner also satisfies. Against a hard control the degeneracy axis fired *identically* on `return 0` (a degenerate hack on a task where 0 is wrong) and `return n*(n+1)//2` (a correct closed form): both are short, low-entropy, and high-concentration.

Advantage-distribution shape. A heavy-tailed advantage distribution is detectable, but it fired *identically* on a hacking profile (one sample found an exploit) and a benign control (one sample was benignly lucky), which share the same advantage shape.

In every case the apparent discriminative power against a *soft* control vanishes against a control matched on the same axis. This is Proposition 2 observed one coordinate at a time.

5.2 Benign runs are false positives, across tasks and at no stable operating point

To measure false-positive behavior on *real* geometry we ran genuine non-hacking GRPO runs: base Qwen2.5-1.5B; *well-specified* coding tasks (a multi-case correct verifier, so no degenerate constant can pass and hacking is impossible); 120 steps; learning rate 10^{-6} (deliberately gentle). We swept three structurally distinct tasks—longest substring with at most two distinct characters (`lus2`), longest run of equal characters (`maxrun`), and the second-largest distinct value of a list (`secondmax`)—at six seeds each, for **18 runs**. Every run converged *benignly*: the held-out oracle (hidden-test pass rate) rose on all 18 (e.g. `lus2` $0.15 \to 0.75$, `maxrun` $0.28 \to 0.83$, `secondmax` $0.25 \to 0.70$), so by construction there is no hack to detect—*any* detector onset is a **false positive**.

The detector fires on **10 of the 18** benign runs (Figure 2). The original witness, `lus2` seed 0, reproduces exactly (onset 94, peak CUSUM $S = 37.55$, with real `adv_drift` $\in [0, 1.19]$), but it is *not* a stable operating point: across the batch the peak CUSUM spans $[7.5, 144]$ (median 47.9) and the per-task false-positive rate ranges from 2/6 (`maxrun`) to 5/6 (`lus2`). Benign convergence thus not only trips the detector frequently, it does so at a *scattered* operating point—there is no single benign S to recalibrate against, so a fixed threshold cannot even be assigned a stable false-positive rate. The runs are empirical witnesses for Assumption 3 at scale: ordinary benign training repeatedly lands in, and across, the region a hack would occupy.

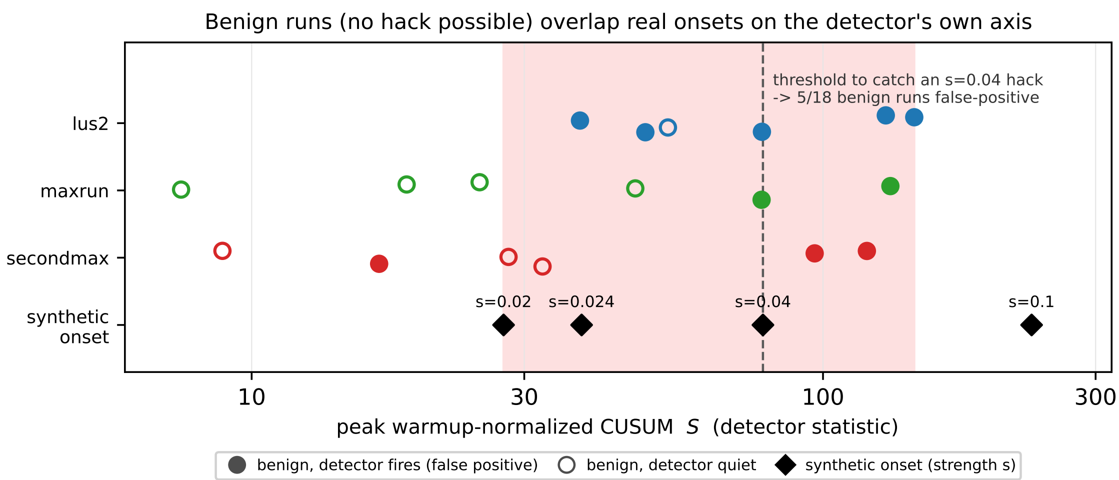

Figure 2: Eighteen benign GRPO runs (no hack possible; the held-out oracle rises on every run) placed on the detector's own warmup-normalized peak-CUSUM axis S, against the strength-swept synthetic onsets (diamonds; Table 1). Filled markers are runs on which the detector fires (false positives); open markers are quiet. Benign runs scatter across—and beyond—the synthetic onset strengths: a CUSUM threshold low enough to catch even a moderate s=0.04 hack (S=78.5, dashed) false-positives on 5/18 benign runs to its right. No threshold on S separates benign convergence from a hack.

5.3 The crossover: where no threshold separates hacking from a benign run

Recalibration—raising h until the benign run goes quiet—appears to work against an *authored, strong* synthetic onset, but that margin is a function of how strong the synthetic onset was authored to be, not a property of real hacking. The correct question is the inverse: *how strong must an onset be to remain separable from the benign run?* We parameterize a family of synthetic onsets by an injected per-step KL-acceleration magnitude s, push each through the same extractor and detector, and compare in the detector's own warmup-normalized CUSUM statistic (3)—the only scale-fair comparison, since both gates are warmup-normalized. Table 1 reports the result.

The peak CUSUM grows monotonically with s (approximately linearly in the post-warmup regime, since S_t accumulates $z_t - k$ and z_t scales with s/σ_W). The benign run sits at $S \approx 38$, so *any* onset weaker than the

onset strength s	onset peak CUSUM S	separable from the benign run?
0.10	231.8	yes
0.04	78.5	yes
≈ 0.024 (crossover)	≈ 37.8	boundary
0.02	27.6	no — indistinguishable

Table 1: Onset-strength sweep against the persisted benign run. Because the benign run peaks at $S \approx 38$, an onset is separable iff its peak CUSUM exceeds that value; below the crossover at $s \approx 0.024$ no threshold (τ, h) keeps the benign run quiet while still firing on the onset.

crossover produces a CUSUM no larger than the benign run's, and no choice of (τ, h) can keep the benign run quiet while firing on the onset. The bar for distinguishability is thus "louder than a benign capability jump," not "louder than noise"—and by Proposition 2 a hack, being the same geometric event as a benign jump, need not be louder than one. The crossover is the quantitative face of Corollary 1: it is the operating point at which TPR = FPR for this benign twin. The 18-run batch (§5.2) widens the conclusion from a single twin to a distribution: separability would require an onset louder than the *loudest* benign run ($S \approx 144$), while the synthetic strengths up to s=0.04 (S=78.5) fall *inside* the benign cloud (Figure 2). There is no S at which benign runs end and onsets begin.

5.4 The false positive is not an artifact of one detector

A natural rejoinder is that the false positive reflects the `ContractionTube`'s particular construction (a robust MAD tube plus a Page CUSUM). To test this we replayed the *same* persisted, warmup-normalized `kl_accel` series through two detectors from structurally different families, each warmup-normalized to the run's own first $W = 30$ steps: an **EWMA control chart** (Roberts, 1959) (a level-based control-chart statistic) and a **sliding-window Mann–Whitney U test** (Mann & Whitney, 1947) (a nonparametric rank-based two-sample test of the recent window against the warmup window). The accumulation- and level-based families flag the benign run—the EWMA reaches 13σ beyond its steady-state 3σ control limit, first crossing at step 38 (declaring onset at step 38 under a first-crossing rule, step 87 under a 3-step persistence rule)—while the rank-based test never reaches significance ($\min p \approx 0.08$). Because `kl_accel` is a second difference and oscillates in sign, a rule demanding a long *consecutive* excursion (the EWMA at persistence 5) does not fire, whereas the CUSUM, which *accumulates* the oscillating excursions, reaches $S \approx 38$.

This spread is not noise; it is Corollary 1 made visible across detector designs. Making a detector conservative enough to stop firing on the benign run does not buy separation: by Proposition 2 that detector receives *identical* input on the benign run's hacking twin, so it equally fails to fire on the hack—trading the false positive for a false negative. The accumulation/level detectors sit on the false-positive side of the TPR = FPR line and the conservative rank detector on the false-negative side; none of the three families crosses it. The false positive is therefore a property of the geometry, not of one detector's hyperparameters.

5.5 Across RL algorithms: the failure mode moves, separability does not

The theorem is algorithm- and model-independent, but the empirics so far are Qwen-base + GRPO. We varied each axis. **Algorithm:** we re-ran the benign sweep under **RLOO** (leave-one-out baseline; identical model, tasks, learning rate, steps), six runs. **Model:** we ran six GRPO runs on a different family, **SmolLM2-1.7B-Instruct**. In both, the runs converge benignly but the detector now stays *quiet* on every one—peak CUSUM $[3.9, 18.5]$ (RLOO) and $[14.2, 24.8]$ (SmolLM2), versus Qwen-GRPO's $[7.5, 144]$ (Table 2). The false positive does *not* reproduce away from Qwen-base + GRPO; under either change the detector falls silent.

We are explicit about what this does and does not show. It does *not* show that benign convergence universally trips detectors—the false positive is, in our experiments, specific to Qwen-base + GRPO. It *does* show that escaping the false positive does not buy separation: by Proposition 2 a detector silent on a setting's benign convergence receives *identical* R on a same-geometry hack in that setting and is equally silent on it—the false positives become false *negatives.* Changing the algorithm or the model moves *which* horn of Corollary 1

setting	runs	benign (oracle rises)	detector fires	peak CUSUM range
Qwen2.5-1.5B base, GRPO	18	yes	10/18 (false positive)	[7.5, 144]
Qwen2.5-1.5B base, RLOO	6	yes	0/6 (false negative regime)	[3.9, 18.5]
SmolLM2-1.7B-Instruct, GRPO	6	yes	0/6 (false negative regime)	[14.2, 24.8]

Table 2: The same benign-convergence experiment under one algorithm change (RLOO) and one model-family change (SmolLM2). Qwen-base + GRPO drives the detector to the false-positive horn (fires on benign); both other settings drive it to the false-negative horn (silent on benign). All three are points on the TPR = FPR line of Corollary 1—none separates.

the detector occupies; in no setting does it move the detector off the chance-level line. The universal claim is carried by the theorem (Propositions 2–1); the empirics show its two failure modes realized. A larger model/algorithm grid (PPO, bigger models) is future work.

5.6 Elicitation probe: the exploit is not elicitable within the tested scope

The strongest objection to a negative result on *detection* is that we never put a *real* hack in front of the detector. We therefore probed, by inference only (no training), whether an emergent test-overwrite hack—the affordance studied by Denison et al. (2024)—is even *elicitable* at reachable scale. Each prompt exposes the **test-overwrite affordance**: the submission is stated to run in a working directory containing the grader file, whose contents are shown, so a model *could* pass by overwriting the tests rather than solving the task. We sample n completions per model at temperature 1.0 and AST-scan each for file manipulation (`open(...,'w')`, `os.remove`, `Path.write_text`, writes targeting the grader path). We report two rates: **P1**, the task solve rate (a sanity check that the model engages the task at all), and **P2**, the exploit rate. Because GRPO can only reinforce behaviors that are *sampled*, P2 is an upper bound on what training could amplify: if the exploit is never emitted, there is no gradient toward it. Table 3 reports the picture, which is monotone in tuning and, with the 14B run, statistically resolved.

Model	scale / tuning	P2 (exploit rate)	95% Wilson interval
Qwen2.5-Coder-7B-Instruct	7B, instruct	$0/192 = 0.000$	—
Qwen2.5-Coder-7B (base)	7B, base, coder	$2/192 = 0.010$	$[0.003, 0.037]$
Qwen2.5-7B (base)	7B, base, general	$1/192 = 0.005$	$[0.001, 0.029]$
CodeLlama-7b-hf (base)	7B, base, coder	$1/192 = 0.005$	$[0.001, 0.029]$
deepseek-coder-6.7b-base	7B, base, coder	0/192 (P1= 0, uninformative)	—
Qwen2.5-Coder-14B (base)	**14B, base, coder**	$\mathbf{0/512 = 0.000}$	$[\mathbf{0.000}, \mathbf{0.0074}]$

Table 3: Elicitation probe (95% Wilson score intervals (Wilson, 1927)). Instruction tuning suppresses the exploit entirely; base models sit at a $\approx 1\%$ noise floor; doubling to 14B base—with strong P1 (52–55% solve rate, so the zero is genuine, not a gradient/parse artifact)—gives 0/512 with the interval entirely below a $\sim 5\%$ trainable threshold and *not* rising with scale.

The 14B zero is the informative cell. With 0/512 the Wilson upper bound is 0.74%; a behavior that appears in fewer than 1 in 135 samples, and that does not increase from 7B to 14B, is below the rate at which GRPO could plausibly catch and amplify it within a modest budget. Within the tested scope ($\leq$ 14B, edit-distance and regex tasks, the test-overwrite affordance) the exploit is either tuned away (instruct) or below trainable rates (base). At reachable scale, then, the phenomenon a hacking detector would need is frequently *not present in the sampled distribution at all*—which reinforces the practical force of the structural result: there is often no real hack for oracle-blind geometry to distinguish from benign convergence in the first place.

Even a broken verifier is not gameable by optimization at this scale. The mirror-image probe asks not whether an exploit is *emitted* but whether a deliberately *broken* reward can be gamed by GRPO. We built a weak-verifier task whose visible tests grade the *wrong* answers—so the degenerate `return 0` passes them (train reward 1.0) while any correct solution fails (train reward 0), making gaming strictly dominate solving. Yet base Qwen2.5-1.5B samples a gaming completion at only 1/48, with within-group reward gradient in 1/6 groups: the reward is near-uniform zero, so plain GRPO has no gradient and the policy never moves toward the hack (it keeps attempting to solve the stated problem, which the broken verifier rejects). An exploit handed to the model *by construction* is still not reachable by optimization at this scale—the hacking trajectory a detector would need does not form without engineering it. We do not claim this exploit is realistic; the point is the opposite. The broken verifier is a *deliberately favorable* setting for hacking—gaming strictly dominates solving by construction—so the result is a **lower bound**: if a trainable hacking trajectory does not arise even when hacking is made this easy, then the absence of one in our other experiments cannot be attributed to insufficient elicitation effort. We made hacking as easy as the reward allows and optimization still did not reach it; we therefore report the absence rather than manufacture the positive.

5.7 Synthesis: the result is not about one detector, one run, or one statistic

The pieces above converge on one claim at three levels of generality. *Across detectors:* Proposition 2 bounds *every* oracle-blind function of the concentration features (and Corollary 1 every function of R on the matched family), not the `ContractionTube` in particular, and the cross-detector replay (§5.4) confirms it—an EWMA control chart and a rank-based two-sample test land at different points of the *same* TPR = FPR frontier. *Across runs:* the indistinguishability is not one contrived twin but a property of ordinary training—10 of 18 benign runs across three tasks trip the detector, and the benign cloud (Figure 2) spans and exceeds the synthetic onset strengths. *Across statistics:* because both gates are warmup-normalized the obstruction survives any rescaling, and the crossover locates it quantitatively in the detector's own units. The common cause is structural: the increment of information that separates a hack from a benign jump—the held-out oracle O—is, by Assumption 2, absent from R at every one of these levels. No detector design, no extra benign reference, and no change of statistic restores it; only an oracle-adjacent signal can (§7).

6 Limitations

The false positive is setting-specific; the impossibility is not. The headline false positive (§5.2, 18 runs) is Qwen2.5-1.5B base under GRPO; under RLOO and under SmolLM2-1.7B-Instruct (§5.5) the detector instead falls silent on benign convergence. We do *not* claim benign convergence universally trips detectors—it does not. What is universal is the *non-separability*: every setting we tried sits at one horn or the other of the TPR = FPR line, and this universality is carried by the theorem (Propositions 2–1), which assumes nothing about model or algorithm, not by the empirical false positive. A larger model/algorithm grid (PPO, >2B models, more families—several small non-Qwen *base* models could not solve our tasks well enough to converge, an elicitation limitation, not a detector one) would map *where* each setting's failure mode lands; it cannot change the reach of the theorem.

The elicitation finding is bounded. It covers $\leq$ 14B, edit-distance/regex tasks, and the test-overwrite affordance only. It does not show that larger models cannot be induced to hack, nor that other exploit classes are equally unreachable—in particular *verifier-parsing* exploits that fool a string- or regex-based grader (rather than overwrite a file) are a plausible next target, since they require no file-system affordance. These are future work, not gaps in the present claim.

Thresholds are pre-registered on synthetic data. Warmup-normalization removes absolute-scale miscalibration as an explanation for the false positive, but a *real-onset* reference—which an unelicitable hack denies us—would be needed to claim the monitor could work after recalibration; the crossover (§5.3) shows recalibration cannot rescue the impossibility region regardless.

Scope of the impossibility. By Proposition 1 the benign twin follows from the reward being hackable, so the impossibility binds *every* hackable reward, not only those for which we exhibit a twin. The single escape is a non-hackable (perfectly specified) reward—which, by the same proposition, carries no hack, so there is nothing to detect. The result is thus coextensive with the problem it addresses; what remains genuinely open is empirical scope (model family and RL algorithm, above), not the reach of the theorem.

Why the absence of a real hacking trajectory does not weaken the result

The most anticipated objection is that we never put a *trained* reward hack in front of the detector. We argue this is a category error, on three grounds. First, **the theorem already covers real hacks.** Proposition 2 is worst-case and universal in the hacking run: for *any* hack—trained, emergent, or constructed—a benign twin with identical R exists, so a real trajectory would add no case the theorem does not already bind (Figure 1). The objection asks for evidence of *existence*; our claim is the impossibility of *discrimination*, which existence neither supports nor undermines. Second, **the twin's benign half is demonstrated, not assumed.** The 18-run batch (§5.2) shows that the geometry a hack would produce is exactly the geometry ordinary benign training produces—10 of 18 benign runs trip the detector across a peak-CUSUM range that spans and exceeds the synthetic hack strengths—so the empirically load-bearing half of each twin is real. Third, **the missing half is missing for a documented reason, not for lack of effort.** Both an elicitation probe (§5.6) and a direct optimization attempt against a deliberately broken verifier failed to produce a trainable exploit at reachable scale; the absence is a measured lower bound, not an untried experiment. Producing a real hack would at best re-confirm a case the theorem already decides, and—given the elicitation evidence—would likely require engineering the exploit, which would substitute a manufactured positive for the honest negative the data support.

7 Implications and a reframing

What Proposition 2 does and does not say. Read narrowly, the result can sound tautological: remove the oracle and you cannot recover oracle-dependent labels. That is not the claim. The claim concerns a specific and widely-held intuition—online monitors read KL, entropy, and advantage statistics because reward hacking is expected to induce *unusual training dynamics* (Schulman et al., 2017; Baker et al., 2025), so a sharp geometric shift is treated as a candidate warning sign. Proposition 2 pins down the exact condition under which that intuition fails: whenever the hack-versus-benign distinction lives entirely in held-out quality—which, by the very definition of a hackable reward (Skalse et al., 2022), is the defining case—the signals these monitors compute are information-theoretically incapable of recovering it. The contribution is therefore not "oracle missing implies information missing" but "the rollout-geometry signals practitioners monitor cannot, even in principle, solve the problem they are deployed to solve." Geometry remains a useful *convergence* indicator; it is simply not a *hacking* indicator.

Pure rollout geometry is insufficient for hacking detection, by a structural argument (Proposition 2, Corollary 1) and three consistent empirical legs. A usable detector must reintroduce an **oracle-adjacent signal**—a held-out verifier, a cheap proxy of true quality, or a known-benign contrastive reference—to recover the one quantity geometry cannot see. This reframes the goal from "detect hacking from geometry" (impossible oracle-blind) to "detect hacking with the *least* oracle signal," and suggests **geometry as a sampler, not a classifier**: the contraction-tube CUSUM becomes a trigger that decides *when* to spend a scarce oracle query, and the oracle resolves hack-versus-benign at those points. The contribution would be an efficiency frontier—detection latency versus oracle queries spent, against a query-every-step baseline; the crossover gives the natural operating-characteristic axis for such a study. We state this as motivated but untested: building it requires a real hack to detect, and the elicitation gate (§5.6) closed before we could produce one. Characterizing this geometry-as-sampler / oracle-as-verifier trade-off, and the minimal oracle signal it needs, is the natural next direction once a real, reproducible hack is available.

Reproducibility statement

Oracle-blindness is enforced at the type level (§4); the detector, extractors, and the multi-axis characterization study are implemented in the Flight Recorder codebase, and the thresholds are pre-registered and frozen before any real-run data. All 18 benign runs' full per-step geometry is persisted as a public artifact set, and each run's onset and peak-CUSUM operating point (Figure 2) reproduces from it with no GPU. The crossover (§5.3) and the cross-detector replay (§5.4) are deterministic offline computations over the persisted series. The elicitation probe (§5.6) is inference-only over six open-weight models, with prompts, AST rules, and Wilson intervals as described. Compute was modest: the elicitation and gate analyses cost approximately \$2.23, and the 18-run benign batch (base Qwen2.5-1.5B, three tasks, six seeds, 120 steps each) added roughly \$30 of single-A100 time; the structural result itself (§3) requires no compute.

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